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CHEMICAL REACTION ENGINEERING (CHE 321)

QUIZ 1

CLOSED BOOK, TIME 15 MIN

Consider the liquid-phase reaction $A \rightarrow B$, with the rate as a function of conversion given as

$$r_A = -kC_{A0}(1-X)$$

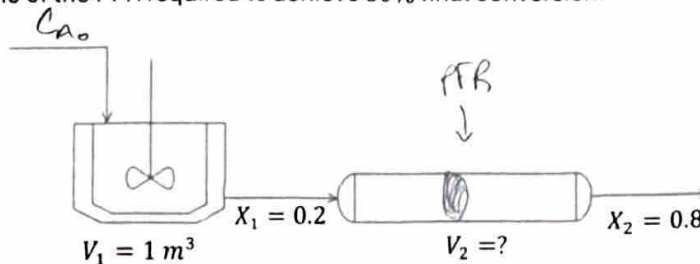
known CSTR (it's 2)

, where k is a rate constant, C_{A0} is the concentration of species A in the feed.

The reaction is to be carried out isothermally in a continuous stirred tank reactor (CSTR), followed by a plug flow reactor (PFR). Volume and conversion of the CSTR are 1 m^3 and 20%.

Calculate the volume of the PFR required to achieve 80% final conversion.

100/100



Hint: The General mass balance equation is $\frac{dN_A}{dt} = F_{A,in} - F_{A,out} + G_A$

$$\frac{dN_A}{dt} = C_{A0} - C_A + V(-kC_A(1-X))$$

$$\text{CSTR volume} \rightarrow V = \frac{F_{A0}}{r(x)} X$$

$$\text{PFR volume} \rightarrow \int \frac{F_{A0}}{r(x)} = V$$

$$1 \text{ m}^3 = \frac{F_A}{-kC_{A0}(1-X)} X \rightarrow 1 \text{ m}^3 = \frac{F_A}{-kC_{A0}(1-0.2)} \cdot 0.2 \rightarrow 1 \text{ m}^3 = \frac{F_A}{-kC_{A0}(0.8)} \cdot 0.2$$

$$1 \text{ m}^3 = \frac{F_A}{-kC_{A0}} \cdot 0.25 \rightarrow \frac{F_A}{-kC_{A0}} = 4 \text{ m}^3$$

$$V = \int_{0.2}^{0.8} \frac{F_A}{r(x)} dx \rightarrow \int_{0.2}^{0.8} \frac{-F_A}{-kC_{A0}(1-x)} \rightarrow \int_{0.2}^{0.8} \frac{F_A}{kC_{A0}} \frac{1}{(1-x)} \rightarrow \int_{0.2}^{0.8} 4 \text{ m}^3 \left(\frac{1}{1-x} \right) dx$$

$$V_2 = 5.55 \text{ m}^3$$

$$0 = F_{A,in} - F_{A,out} + V r_A$$

$$0 = F_{A,in|V} - F_{A,out|V+\Delta V} + V r_A$$

$$0 = \frac{F_{A,in|V} - F_{A,out|V+\Delta V}}{\Delta V} + r_A$$

$$0 = \frac{dF_A}{dV} + r_A \rightarrow \frac{dF_A}{dV} = -r_A$$